

WEEK 7 CROSS-TRACK INTEGRATION & HANDOFF RECORD

Project: HealthConnect Clinic Appointment No-Show Prediction

Track: Data Science

Collaborating Track: Data Analytics

Week: 7 - Testing, refinement & validation

Integration type: Data Science ↔ Data Analytics model validation and findings handoff

1. Purpose of the Handoff

This document provides formal evidence of the Week 7 cross-track integration between the Data Science and Data Analytics tracks. The purpose of the collaboration was to exchange validated analytical and modelling findings and use those findings to test whether the Week 6 candidate model required further refinement. The collaboration was therefore not limited to communication. Findings received from the Data Analytics track were incorporated into the Data Science testing process, used to motivate a specific feature refinement, tested against the existing candidate model, and subsequently evaluated to determine whether the refinement should be retained. The Data Science results were also shared back with the Data Analytics track to provide evidence of how the analytical findings were reflected in the predictive modelling workflow.

2. Cross-track dependency

Data Analytics → Data Science

The primary dependency was:

Validated analytical findings → Data Science feature review, model testing and refinement

The Data Analytics track provided findings concerning variables associated with appointment no-show behaviour, particularly:

- Booking lead time,
- Previous no-show history,
- Distance to the clinic,
- Appointment characteristics,

- The relationship between booking lead time and previous no-show history.

These findings were used by Data Science to determine which aspects of the Week 6 model required further investigation during Week 7.

Data Science → Data Analytics

The Data Science track provided the Data Analytics track with:

- The retained model feature set,
- Target-variable definition,
- Final candidate model performance,
- Confusion-matrix results,
- Error-analysis findings,
- Segment-level model performance,
- Results of the Week 7 feature-refinement experiment,
- The final decision regarding the tested refinement.

This created a two-way information flow between the two workstreams.

3. Information received from Data Analytics

The Data Analytics findings identified booking lead time and previous no-show history as particularly important variables for understanding no-show behaviour. The analysis showed a strong ordered relationship between booking lead time and no-show rates.

The observed no-show rate increased from approximately:

- 29.47% for appointments booked 0-7 days in advance
- to 71.36% for appointments booked 46-60 days in advance.

This finding indicated that booking lead time required further investigation during the Data Science testing stage. Previous no-show history was also identified as an important variable and supported its continued inclusion in the predictive model. The Data Analytics

findings therefore provided a specific analytical basis for examining the representation of booking lead time in the Week 7 model.

4. Information provided by Data Science

The Data Science track provided the following modelling information to the Data Analytics track.

4.1 Final candidate model

The retained candidate was the Week 6 Refined Logistic Regression model.

The model used 13 features:

- Gender
- Age
- age_group
- appointment_type
- appointment_day
- booking_lead_days
- booking_lead_group
- previous_appointments
- previous_no_shows
- previous_no_show_group
- distance_to_clinic_km
- distance_group
- lead_no_show_interaction

The target variable was:

no_show

1 = No-Show

0 = Attended

Cancelled appointments were excluded from the binary modelling task.

5. Candidate model performance shared

The retained candidate model achieved the following test-set performance:

Metric	Result
Accuracy	61.50%
Precision	60.87%
Recall	69.28%
F1 Score	64.80%
ROC-AUC	66.47%

The confusion matrix was:

[[247, 216],
[149, 336]]

This corresponds to:

Classification outcome	Count
True Negative	247
False Positive	216
False Negative	149
True Positive	336

The results were shared with Data Analytics to provide the modelling perspective alongside the descriptive analytical findings.

6. How the Data Analytics findings influenced Data Science

The Data Analytics findings directly influenced the Week 7 modelling investigation.

Previous no-show history

The analytical finding that previous no-show history was relevant supported retaining:

previous_no_shows

previous_no_show_group

in the model.

It also supported the interaction feature introduced during Week 6:

lead_no_show_interaction

Booking lead time

The strong ordered relationship between booking lead time and no-show behaviour motivated Data Science to investigate whether the existing representation of booking lead time could be improved. Rather than assuming that the existing feature representation was sufficient, a targeted refinement was therefore tested in Week 7.

Reminder information

Although reminder-related variables may be relevant to the broader HealthConnect operational analysis, reminder_sent was not reintroduced into the predictive model. The variable was excluded because its availability at prediction time could create a potential data leakage or prediction-time availability problem. This distinction was communicated so that the Data Analytics and Data Science workstreams remained aligned while maintaining appropriate modelling controls.

7. Collaborative testing activity

The main Week 7 cross-track testing activity was:

Testing whether the validated Data Analytics finding on booking lead time justified an improved representation of booking lead time in the predictive model. Based on the analytical evidence, Data Science introduced an additional logarithmic transformation:

```
df["booking_lead_log"] = np.log1p(df["booking_lead_days"])
```

The feature was added to the Week 7 modelling feature set and evaluated against the retained Week 6 candidate. The purpose was to determine whether a non-linear representation of booking lead time would improve predictive performance.

8. Results of the collaborative refinement test

The Week 7 refinement was compared with the retained Week 6 candidate.

Metric	Week 6 Candidate	Week 7 Refinement	Change
Accuracy	61.50%	61.50%	0.00 pp
Precision	60.87%	60.91%	+0.04 pp
Recall	69.28%	69.07%	-0.21 pp
F1 Score	64.80%	64.73%	-0.07 pp
ROC-AUC	66.47%	66.52%	+0.05 pp

The changes were very small.

The refinement also did not produce consistent improvement across the booking lead-time segments. Therefore, the additional `booking_lead_log` representation was not adopted.

9. Cross-track decision

The joint evidence resulted in the following decision:

Retain the Week 6 Refined Logistic Regression model as the current candidate model. The `booking_lead_log` refinement was rejected because it did not produce a meaningful or consistent improvement in overall or segment-level performance. This was an important Week 7 outcome because the purpose of the cross-track integration was not to force an improvement into the model, but to determine whether the analytical finding justified a modelling change. The evidence indicated that the existing representation was sufficient for the current stage of the project.

10. What changed as a result of the integration?

The cross-track integration resulted in several concrete changes to the Week 7 work:

1. Data Analytics findings were incorporated into the Data Science testing strategy,

2. Booking lead time was identified as a specific area requiring targeted model testing,
3. A new booking_lead_log feature was developed and tested,
4. The refinement was evaluated against the retained Week 6 candidate,
5. Segment-level performance was considered alongside overall metrics,
6. The new feature was rejected because the evidence did not support retaining it,
7. The Week 6 candidate model was retained,
8. The Data Science findings were communicated back to Data Analytics.

Therefore, the collaboration resulted in an actual modelling experiment, evidence-based decision and documented model outcome, rather than communication alone.

11.Evidence of integration

The following artefacts provide evidence of the cross-track activity:

Evidence	Purpose
Data Analytics validated findings	Analytical basis for model testing
Data Science model feature list	Shows the features evaluated
Week 7 booking_lead_log feature	Evidence of targeted refinement
Week 6 vs Week 7 model comparison	Evidence of testing
Segment-level booking lead analysis	Evidence of refinement validation
Final model metrics	Evidence of model evaluation
Confusion matrix	Evidence of error evaluation

Week 7 re-test results	Evidence that the retained candidate was confirmed
Cross-track handoff document	Evidence of information exchange between tracks
Week 7 notebook	Technical evidence of the testing and refinement process

12. Overall cross-track outcome

The Week 7 collaboration strengthened the connection between Data Analytics and Data Science workstreams. The Data Analytics track provided evidence identifying booking lead time and previous no-show history as important factors in appointment no-show behaviour. The Data Science track translated these findings into a specific modelling investigation and tested whether an improved representation of booking lead time would strengthen the candidate model. The additional feature did not produce meaningful improvement and was therefore rejected.

The outcome demonstrates that the cross-track findings were:

Received → Interpreted → Implemented → Tested → Evaluated → Documented

The collaboration therefore contributed to evidence-based model refinement while avoiding unnecessary changes to the candidate model.

13. Status after cross-track testing

Following the integration and testing activity:

Retained Candidate: Week 6 Refined Logistic Regression

Tested Refinement: booking_lead_log

Refinement Decision: Rejected

Current Model Status: Suitable for continued testing and development

Production Status: Not production-ready

Further validation, including additional validation data, threshold analysis, probability calibration and broader integration testing, remains necessary before any operational deployment could be considered.

14. Handoff to the Data Analytics track

The Data Science track therefore formally hands back the following information to Data Analytics:

1. The Week 6 Refined Logistic Regression remains the retained candidate,
2. Booking lead time remains an important modelling variable,
3. Previous no-show history remains an important modelling variable,
4. The booking_lead_log refinement was tested but rejected,
5. The candidate model achieved 61.50% accuracy, 60.87% precision, 69.28% recall, 64.80% F1 and 66.47% ROC-AUC,
6. The candidate produced 216 false positives and 149 false negatives,
7. Segment-level testing indicates that shorter booking lead-time groups require continued attention,
8. No additional feature change is justified from the Week 7 booking lead-time refinement test,
9. The model remains suitable for further testing and development but is not production-ready.
10. These findings should be considered in the final HealthConnect integration and Week 8 validation activities.

15. Final cross-track statement

The Week 7 Data Science and Data Analytics collaboration resulted in a documented exchange of analytical and modelling findings and a concrete modelling test. The Data

Analytics findings directly influenced the Data Science investigation of booking lead time. The resulting feature refinement was tested using the HealthConnect candidate model and subsequently rejected because the evidence did not demonstrate meaningful improvement. The final outcome was therefore not simply an exchange of information, but a traceable cross-track workflow in which analytical evidence informed modelling, modelling results were returned to the analytical workstream, and the combined evidence informed the final modelling decision.